

COMP3018: Set Exercises; Human-Robot Interaction (HRI)

Cultural Differences and Probabilistic Modelling in Human-Robot Interaction

Contents

Words-To-Use:	1
TODO:	2
1- Task (1): Cultural Differences and HRI Design	3
1.1. Cultural Differences in the Acceptance of Robots (Kaplan, 2004)	3
1.1.1 Western Society (The Frankenstein Syndrome)	3
1.1.2 Eastern Society (Technology Taming and Animism)	3
1.1.3 Implications for HRI Design.	3
1.2. African Cultural Factors Influencing HRI Design	4
1.2.1 Ubuntu Philosophy and Communal Identity:	4
1.2.2 Power Distance and Hierarchical Norms:	4
1.2.3 Oral Tradition and Multimodal Communication:	5
1.2.4 Infrastructure and Access Constraints:	5
1.3 Regional Design Traits (Appearance and Behaviour)	5
1.4 Adapting Design Patterns for Sociality (Kahn et al., 2008)	6
1.5 Summary of Cultural Design Implications	7
2- Task (2): POMDPs in Human-Robot Interaction	8
2.1 The Role of POMDPs in Trust, Cooperation, Coordination, and Collaboration	8
2.2 Uncertainty, Belief States, and Decision-Making	8
2.3 Challenges of Trust Modelling and the POMDP Response	9
2.4 Proposed Neuro-Symbolic POMDP Model: Neo Robot with OpenAI Cognitive Architecture	9
2.5 Ethical and Social Implications	11
References	12
Task (1)'s	12
Task (2)'s	12
Appendices	14
Appendix A: AI Declaration	14

Words-To-Use:

- ☐ **Lecturer's Top Insight:** Be a Reviewer; don't just argue opinions. Validate every critique with evidence from the literature to ensure it is scientific, not personal.
- ☐ INTEGRATE ROBOTIC LaTeX DIAGRAM
- ☐ 3018-CW/learning/lectures/5 [] - utilise/lecture.md (**Task 1 insight!!**)
- ☐ 3018-CW/learning/lectures/6 [] - utilise/lecture.md (**General cw insight**)
- ☐ use `whom`

- ☐ talk about novelty effect in lecture 6
- ☐ most-{something} {something}
- ☐ would-be
- ☐ despite
- ☐ contravened — [] humanlike
- ☐ watchword
- ☐ e.g. 1- , 2-, ...
- ☐ despite x something y
- ☐ and indeed, the robot will...
- ☐ and therefore, x thus {does, e.g. *feeds* y continuously throughout the process wherein...
- ☐ as now-{something} the x thus does y continuously throughout the process wherein it does z
- ☐ Whilst: only used at the start of a sentence
- ☐ Whilst this is true, x may be inclined to {x} based on...
- ☐ and thus *x* therefore...
- ☐ is indeed...
- ☐ and indeed...
- ☐ however, insofar as
- ☐ thereof: of the thing just mentioned
- ☐ herein: in this document
- ☐ Note
- ☐ Approach
- ☐ within
- ☐ wherein: in which
- ☐ regarding, in regard to, etc
- ☐ likelihood
- ☐ thereof:
- ☐ infer: conclude from reasoning
- ☐ init: initialise
- ☐ use: use
- ☐ via: through
- ☐ wherein: in which
- ☐ indeed: in fact

TODO:

- ☐ verify page numbers are correct
- ☐ hit word-count*0.1 limit allowance across report
- ☐ utilise lecture teachings in lec 5-6 etc
- ☐ review papers: pros or limitations
- ☐ indeed 10% word-count allowance
- ☐ peer-reviewed or conference papers
- ☐ very good LaTeX visualisations
- ☐ uses 3003-report feedback??
- ☐ In this section, you should focus on providing enough description of the supervised learning, neural network, and naïve Bayes models.
- ☐ Do not assume the reader knows the basics. Dedicate specific paragraphs to explicitly defining the algorithms and the broader category (Supervised Learning) before diving into your implementation.
- ☐ Then, refer to some studies that have utilised neural networks and naïve Bayes models in your area using the selected database
- ☐ Ensure your literature review in the introduction explicitly cites papers that use your specific dataset (or very similar ones), establishing a clear baseline before you begin

1- Task (1): Cultural Differences and HRI Design

~1,750-word | 40% weighting | Questions 1.1-1.5

1.1. Cultural Differences in the Acceptance of Robots (Kaplan, 2004)

Kaplan’s 2004 identification of East-West fundamental societal divergence is rooted in observation as follows: **“culture affects the way technology is perceived and, reciprocally, technological evolution shapes culture in particular ways”** (Kaplan, 2004, p. 465); i.e. the cultural (habits), theological (religious), and mythological narrative of each region shapes the societal meta-layer (*the underlying philosophical-and-theological framework controlling robot integration for each culture*); this underpins robot acceptance.

1.1.1 Western Society (The Frankenstein Syndrome)

Western culture persistently has viewed the creation of human-like (humanoid) entities slightly suspiciously. Kaplan (2004) identifies this as the “Frankenstein Syndrome”: a culturally-filtered conviction wherein “any artificially created humanoid will necessarily turn against its creator” (p. 475).

This anxiety traces to the West’s distinction between nature and culture, which posits “no place for hybrids” in such classifications (Kaplan, 2004, p. 470). The Western cultural narrative therefore frames the humanoid robot as a challenge to human specificity (p. 476), and thus, a transgression against the natural order; as a result, Western societies have historically envisioned robotic development towards industrial, non-anthropomorphic (instrumentalist, TODO EXPAND) applications wherein the machine remains a *tool* rather than a would-be social entity. Kaplan further notes the concept of “narcissistic shields” (*the psychological defence mechanisms protecting human exceptionalism*) (p. 478), whereby Westerners psychologically distance themselves to manage the discomfort of encountering machines that erode the human-robot distinction.

1.1.2 Eastern Society (Technology Taming and Animism)

Japanese culture: exhibits a fundamentally different ontological (*the definition of what counts as a ‘being’*) stance. Kaplan (2004) traces this to the Shinto tradition, wherein the rigid Western boundary between animate and inanimate is dissolved in favour of a “continuous network of beings” (p. 470). In this view, humanoid robots are not perceived as transgressions but instead as natural extensions.

Furthermore, Kaplan discussed the cultural mechanism of “technology taming” i.e. a recurring historical pattern wherein foreign technologies are domesticated via integration into existing cultural frameworks (p. 467). This ethos aligns with the *kata* tradition of formalised practice, where repetition leads to “maximum stability” (p. 470). The popular *Astro Boy* manga franchise (p. 466) exemplifies this domestication narratively: the robot is cast not as a Frankensteinian threat, but as a heroic companion (p. 466). Kaplan references the Amaterasu myth (p. 469) to argue that Japanese cosmology fundamentally lacks the creator-vs-creation antagonism (the inherent transgression of usurping divine privilege) that underpins Western technophobia, saying simply that “in Japan, no gods created human beings” (p. 476). This cultural openness persists into the contemporary era; as the lecturer observed, Japanese society remains “fine to leave people more with robots” and is “probably less sensitive to risks that might appear from robots” than the West (Lecture 5).

1.1.3 Implications for HRI Design.

The divergent cultural framings dictate interaction design profoundly within HRI; whilst Westerners predominantly prefer robots that maintain clear machine identity markers, thereby preserving the ‘narcissistic shield’ (Kaplan, 2004, p. 478), Eastern users instead welcome humanlike anthropomorphic features that align with animistic expectations (expectations wherein all entities, including artificial ones, are regarded as being a spiritual essence); indeed, Lim, Rooksby and Cross (2021, p. 1321) observed this contrasting preference, confirming that culture significantly influences the acceptance of robotic morphology as Korean participants envisioned human-like (humanoid) companion robots; US participants instead preferred machine-like functional agents.

Whilst this is true, a Western robot-designer may be inclined to impose universal proxemic standards (standards within culturally-defined personal-space boundaries) based on their own cultural norms; however, the approach is fundamentally flawed as people from different cultures have demonstrably different preferences with respect to proxemics, such that what is normal for one may constitute a violation for another (Joosse, Lohse and Evers, 2014, p. 1). Regarding specific regions, Japanese normally demand larger personal-space buffers and non-tactile greeting protocols, whereas Mediterranean cultures tolerate closer approach distances (Joosse, Lohse and Evers, 2014, p. 2); therefore: a culturally-calibrated model feeds local boundaries continuously throughout the process whereby it calculates approach vectors, as failing to respect these bounds means user expectations are contravened, effectively alienating the would-be companion.

Designers must also account for the novelty effect (Lecture 6); the most-prominent factor wherein users for whom the robot represents an entirely novel stimulus exhibit inflated acceptance ratings. Despite the robot representing something novel (fictional), initial engagement masks genuine preferences, and indeed, the novelty effect has conventionally been framed as a source of noise wherein inflated initial ratings obscure the user’s authentic response (Smedegaard, 2019, p. 412). As now-established, the novelty effect will thus feed skewed data continuously throughout the process as it inflates initial ratings, increasing the likelihood of confounding true cultural preference, with slight unfamiliarity thereof. The watchword herein is indeed cultural relativism: e.g. 1) designers must use culturally-calibrated models to init interaction protocols via appropriate channels; and 2) no universal design policy can accommodate these fundamentally different ontological (real-being) commitments. Researchers cannot infer permanent acceptance from early data and instead, the system must evaluate engagement over time.

1.2. African Cultural Factors Influencing HRI Design

□ DONE?

Kaplan’s (2004) Abstract confines the analysis to the East-West axis, framed explicitly as an inquiry into whether robots are “perceived in the same manner in the West and in Japan”, however a complete account of cultural factors in HRI must address the African context, wherein distinct philosophical and socio-structural dimensions shape the robot’s relational acceptance.

The following factors are well-established in African countries, and thus likely should be integrated into the design of HRI systems deployed therein:

1.2.1 Ubuntu Philosophy and Communal Identity:

Ubuntu: a Southern African philosophical principle that “a person is a person through other persons” (Metz, 2007, p. 323). Whereas Western HRI design typically foregrounds individual user experience, Ubuntu-driven design would prioritise communal benefit and relational harmony. A robot operating within an Ubuntu-oriented society should therefore be designed to address the *group* rather than the individual, facilitating collective decision-making and shared resource access (Metz, 2007, p. 324). This contrasts with the individualised personal-assistant paradigm prevalent in Western HRI.

1.2.2 Power Distance and Hierarchical Norms:

- Cirasa and Conti’s (2025, p.7) scoping review identifies Power Distance i.e. a dimension of Hofstede’s cultural dimensions framework (a foundational but non-peer-reviewed model just discussed): a key factor shaping robot interaction norms in HRI, many African societies score high on this dimension, wherein hierarchical authority structures govern social interaction.

In regards to HRI this suggests that robots interacting with users across different social strata () must modulate their behaviour accordingly: deferential language and posture when addressing elders or authority figures, and indeed a more directive interaction style when assisting in contexts where the robot is perceived as an institutional representative (Cirasa and Conti, 2025, p.7). Failing to encode these hierarchical norms risks contravening deeply held social expectations, thereby undermining trust.

1.2.3 Oral Tradition and Multimodal Communication:

- African cultures have historically privileged oral knowledge transmission over written documentation (Winschiers-Theophilus and Bidwell, 2013). This directly implicates interaction modality, i.e. voice-driven, narrative-based interfaces using speech processing and prosodic features such as pitch and MFCCs (Lecture 3) — may achieve higher engagement than text-heavy GUI paradigms. Furthermore, as Lecture 2 established that “65% of communication is non-verbal,” gestural and paralinguistic channels become critical design considerations for African contexts wherein oral expressiveness is culturally normative.

1.2.4 Infrastructure and Access Constraints:

- Despite rapid technological growth, many African regions face infrastructure limitations including intermittent connectivity and limited access to high-specification hardware (Wyche and Steinfield, 2016). HRI systems deployed in these contexts must therefore be robust to connectivity loss, operable on low-power devices, and designed for shared rather than personal ownership — aligning with the communal ethos of Ubuntu.

1.3 Regional Design Traits (Appearance and Behaviour)

- DONE?

The cultural factors have been identified above and dictate distinct morphological and behavioural traits to maximise acceptance in the respective regions.

Figure~1 illustrates how these frameworks distinctly approach robot strategies.

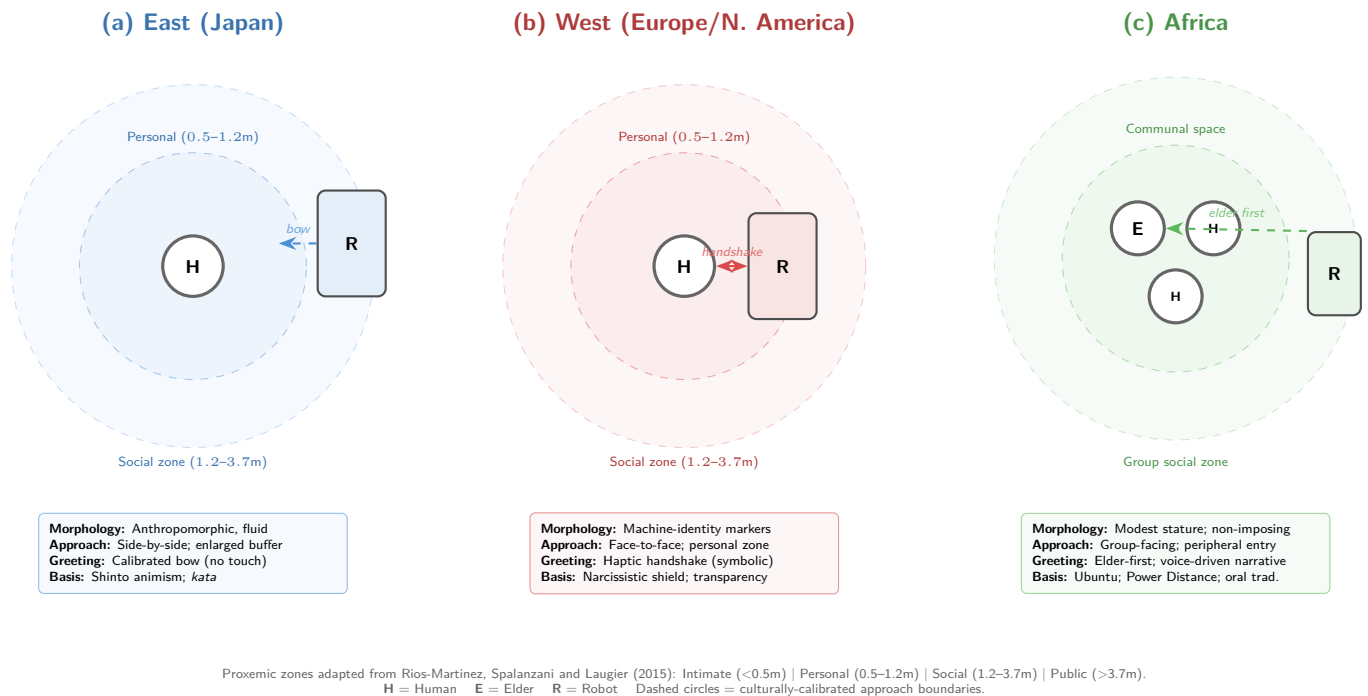


Figure 1: Culturally-adapted proxemic zones and robot approach strategies across three regional paradigms. The robot’s spatial positioning, greeting modality, and morphological design are calibrated to the cultural frameworks identified in Sections 1.1–1.2 (Rios-Martinez, Spalanzani and Laugier, 2015; Kaplan, 2004; Cirasa and Conti, 2025).

(a) **The East (Japan).** Because Shinto animism dissolves the natural/artificial boundary (Kaplan, 2004),

anthropomorphic or highly expressive aesthetic traits are welcomed. However, to align with the *kata* tradition of harmonious form, the robot’s movements must be fluid and graceful rather than purely functional. Behaviourally, the robot should adopt a “side-by-side” cooperative posture rather than an imposing face-to-face stance, reflecting Japanese non-tactile proxemic norms requiring larger personal-space buffers (Lecture 4). Whilst this is true, a designer trained within Western conventions may be inclined to calibrate these buffers using Western proxemic data, which would systematically underestimate the spatial sensitivity of Japanese users. As Lecture 2 establishes, *haptics* (deliberate physical communication) has beneficial effects primarily within the same social group — hence physical touch is replaced by proxemic attentiveness, maintaining Hall’s personal zone (0.5–1.2m).

(b) The West (Europe/North America). To avoid triggering the Frankenstein Syndrome and to respect the “narcissistic shield” (Kaplan, 2004, p. 478), Western robots should possess functional, machine-like rather than overtly humanlike aesthetic markers (e.g., visible joints, metallic chassis) to clearly signal their artificiality, thereby preventing descent into the uncanny valley (Mori, 1970). Behaviourally, they must exhibit extreme transparency, explicitly stating their operational reasoning to alleviate fears of autonomous transgression. The handshake serves as a *symbolic gesture* (a movement with a culturally agreed-upon meaning) combined with *haptics* (Lecture 2), albeit calibrated to the Mediterranean versus Northern European proxemic distinction (Lecture 4).

(c) Africa. Informed by Ubuntu and high Power Distance (Cirasa and Conti, 2025), an African-deployed robot should possess a modest physical stature to avoid perceived challenges to human hierarchical authority. Behaviourally, it must be group-facing rather than dyadic, utilising a warm, highly expressive vocal synthesiser capable of rendering the rich prosodic variations (pitch, tone) necessary for an oral-tradition society (Lecture 3; Winschiers-Theophilus and Bidwell, 2013). Furthermore, as Lecture 2 established that “65% of communication is non-verbal,” gestural and paralinguistic channels — *beat gestures* (rhythmic hand movements accentuating speech rhythm) and *iconic gestures* (movements visually representing the subject) — become critical design considerations.

1.4 Adapting Design Patterns for Sociality (Kahn et al., 2008)

Kahn et al. (2008) identify eight “psychological benchmarks” for social robots, noting the design space is “likely under-described” (p. 100). These patterns must be regionally adapted using the traits established in Section 1.3:

Pattern 1: Initial Introduction (Kahn et al., 2008, p. 100).

- *(a) East:* Rather than tactile handshakes, the robot must initiate interaction with a calibrated bow, as “in Japan it’s very considered impolite if you break the personal distance or space and try to touch somebody” (Lecture 4). The bow angle should parametrically encode social hierarchy recognition.
- *(b) West:* The *Initial Introduction* can incorporate the handshake as a tactile greeting, albeit with sensitivity to the Mediterranean (closer) versus Northern European (distant) proxemic distinctions (Lecture 4).
- *(c) Africa:* Reflecting Ubuntu’s communal orientation, the introduction must address groups rather than individuals. Encoding Power Distance norms, the robot must always greet the eldest or most-senior member first (Cirasa and Conti, 2025).

Pattern 2: Personal Interests (Kahn et al., 2008, p. 101).

- *(a) East:* The robot’s backstory can elaborately integrate into the animistic expectation of objects possessing a “spirit” or character (Kaplan, 2004).
- *(b) West:* Self-disclosure must be transparently mechanical, framing its “interests” around its programmed purpose to reinforce the user’s ontological comfort — supporting Theory of Mind mechanisms (Lecture 1) whereby the robot must infer the user’s mental state.
- *(c) Africa:* This pattern should be realised via storytelling and voice-based dialogue. The robot must establish *joint attention* (the ability of multiple agents to focus on a shared reference point) via structured eye gaze to maintain narrative authority (Lecture 2; Lecture 3).

Pattern 3: Recovering From Mistakes (Kahn et al., 2008, p. 102).

- (a) *East*: The robot should employ indirect acknowledgement strategies that preserve social harmony, avoiding direct self-criticism that may cause discomfort via loss of face.
- (b) *West*: The robot should explicitly explain its error and how it will correct it, prioritising transparency to mitigate underlying technophobia.
- (c) *Africa*: Error recovery must be strictly calibrated to social rank. An unacknowledged robot error affecting an elder constitutes a severe social violation; recovery must involve a disproportionately deferential apology to seniors, leveraging posture and movement (Lecture 2) to express contrition, thereby protecting the user’s social face within the community.

1.5 Summary of Cultural Design Implications

Table 1: Cultural factors and corresponding HRI design adaptations.

Region	Key Cultural Factor	Design Adaptation
East (Japan)	Shinto animism; non-tactile norms; <i>kata</i> tradition	Bowing protocols; enlarged personal-space buffers; indirect error recovery
West	Frankenstein Syndrome; individualism; narcissistic shields	Machine-identity markers; handshake-ready; transparent self-disclosure
Africa	Ubuntu; high Power Distance; oral tradition	Communal greetings; elder-first hierarchy; voice-driven narrative interfaces

As Kahn et al. (2008) conclude, effective HRI must be “compelling as a lived experience” (p. 104) — and what constitutes a compelling experience is, as demonstrated above, inseparable from the cultural context thereof.

2- Task (2): POMDPs in Human-Robot Interaction

□ this needs to change as I am going to incorporate an AI API into the robot's system

2.1 The Role of POMDPs in Trust, Cooperation, Coordination, and Collaboration

A Partially Observable Markov Decision Process (POMDP) extends the standard Markov Decision Process (MDP) — as studied in COMP3003 — by relaxing the assumption of full state observability. Whilst an MDP assumes the agent has direct access to the true environment state, a POMDP acknowledges that in most real-world HRI scenarios, the robot can only *infer* the true state via noisy, incomplete observations (Kaelbling, Littman and Cassandra, 1998). Formally, a POMDP is defined by the 7-tuple $\langle S, A, T, R, \Omega, O, \gamma \rangle$, where S is a finite set of states, A the available actions, $T(s, a, s') = P(s' \mid s, a)$ the transition function, $R : S \times A \rightarrow \mathbb{R}$ the reward function, Ω a finite set of observations, $O(s', a, o) = P(o \mid s', a)$ the observation function, and $\gamma \in [0, 1)$ the discount factor.

To understand the POMDP's utility, one must strictly differentiate interaction paradigms as defined in Lecture 1. **Coexistence** involves agents sharing an environment but completing different tasks, requiring only fully-observable physical states to avoid collisions. **Cooperation** involves a shared workspace and complementary tasks. However, true **collaboration** demands a shared workspace and the *exact same shared goal* (Lecture 1). In collaboration, the robot must continuously align its actions with the human's unobservable mental states — trust, intent, cognitive load. Because the robot is fundamentally blind to these latent variables, the POMDP's belief state b becomes the computational prerequisite for graduating from mere coexistence to true collaboration. The POMDP therefore provides the principled mathematical framework for modelling trust, coordinating joint actions under uncertainty, and enabling genuinely collaborative human-robot teams (Chen et al., 2020). Indeed, Nikolaidis et al. (2017, p. 619) demonstrate this empirically via a “Bounded-Memory Adaptation Model” wherein the robot maintains a POMDP over the human's latent type and adapts its policy accordingly — showing that mutual adaptation via belief-space planning significantly outperforms fixed interaction strategies.

2.2 Uncertainty, Belief States, and Decision-Making

Because the true state is hidden, the POMDP agent maintains a **belief state** b : a probability distribution over all possible states S , where $b(s)$ represents the agent's subjective probability that the environment is in state s , such that $\sum_{s \in S} b(s) = 1$. After taking action a and receiving observation o , the belief state is updated via **Bayesian filtering**:

$$b'(s') = \eta \cdot O(s', a, o) \sum_{s \in S} T(s, a, s') \cdot b(s)$$

η is a normalisation constant ensuring $\sum_{s'} b'(s') = 1$. This update rule captures the core epistemic challenge of HRI: the robot must continuously revise its model of the human's internal state as new — and potentially contradictory — evidence arrives. As Lecture 3 discussed regarding affective computing, the robot utilises descriptors (e.g., pitch, MFCCs, zero-crossing rate) to extract observations from the human's behaviour, and thus *feeds* these into the belief update process whereby each observation incrementally refines the robot's understanding.

The belief state b therefore serves as a **sufficient statistic** for the entire interaction history — it compresses all past actions and observations into a single probability vector, enabling decision-making without storing the full trajectory (Kaelbling, Littman and Cassandra, 1998). This is indeed analogous to how a social robot must infer a human's emotional state from noisy multimodal cues (facial expressions, vocal prosody, posture) rather than accessing the “ground truth” of their feelings directly (Lecture 1; Lecture 3).

2.3 Challenges of Trust Modelling and the POMDP Response

Trust is a latent psychological variable — it cannot be directly measured, only inferred from observable behavioural indicators. Lee and See (2004, p. 54) formally define trust as “the attitude that an agent will help achieve an individual’s goals in a situation characterised by uncertainty and vulnerability” — a definition that foregrounds precisely the latent, goal-dependent nature that necessitates probabilistic modelling. Three core challenges arise. First, **the measurement problem**: observations such as task compliance rate, response latency, gaze direction (Lecture 2), and verbal affirmations are noisy proxies. Hancock et al.’s (2011, p. 520) meta-analysis of 29 empirical studies confirms this difficulty, finding that robot performance-based factors exhibit the strongest correlation with trust (mean $r = +0.26$), yet even these explain only modest variance — a user may comply with a robot’s suggestion despite low trust (e.g., due to time pressure), or indeed refuse despite high trust (e.g., due to task complexity). Second, **temporal dynamics**: trust evolves non-linearly — it builds slowly through consistent performance but degrades rapidly after errors — an asymmetry empirically confirmed by Desai et al. (2013, p. 256), who found that a single autonomous failure eroded trust more than three consecutive successes could rebuild it. The *Recovering From Mistakes* pattern (Kahn et al., 2008, p. 102) is therefore critical: a robot that acknowledges and corrects errors can arrest trust decay, whereas one that ignores failures risks appearing “aggressive” (Lecture 4). Third, **computational intractability**: solving POMDPs exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987), as the belief simplex is continuous even with finite $|S|$. Practical HRI applications therefore utilise approximate solvers — point-based methods such as PBVI (Pineau, Gordon and Thrun, 2003) or online Monte Carlo tree search such as POMCP (Silver and Veness, 2010) — to achieve tractable real-time planning.

The POMDP addresses these challenges by encoding trust as a hidden state variable, observations as probabilistic signals thereof, and actions as trust-modulating strategies (Chen et al., 2020), and thus *trust* cannot be treated as a static binary variable but must be modelled as a continuously-evolving distribution. The belief state thus provides the robot with a principled estimate of trust that is continuously refined, rather than a brittle threshold-based heuristic (a rigid binary switch wherein the subtle fluctuations of trust are ignored).

2.4 Proposed Neuro-Symbolic POMDP Model: Neo Robot with OpenAI Cognitive Architecture

To concretise this framework, I propose a neuro-symbolic POMDP model for a Neo (Pepper) humanoid robot augmented with an OpenAI multimodal API, deployed as an elderly medication-adherence assistant. The fundamental insight is this: large language models (LLMs) and vision-language models (VLMs) are powerful observation extractors — capable of parsing the once-prohibitive complexity of unstructured human behaviour into structured probabilistic assessments. However, LLMs are inherently stateless; each API call is independent, with no temporal memory of prior interactions. They are also stochastic and prone to hallucination — generating plausible but factually incorrect inferences (Ji et al., 2023). Wrapping the OpenAI API inside a POMDP belief state $b(s)$ therefore provides the mathematically rigorous temporal scaffold that the LLM alone lacks: a continuously-updated probabilistic model of the human’s latent states (Trust, Cognitive Load) that persists across the full interaction history. This neuro-symbolic paradigm — wherein a neural subsystem handles perception whilst a symbolic subsystem governs reasoning — represents what Garcez and Lamb (2023, p. 12389) term the ‘third wave’ of AI. In the robotics domain specifically, Ahn et al. (2022) demonstrated via their SayCan framework that LLMs can ground abstract language commands in physical robotic affordances, establishing the feasibility of LLM-directed action selection. However, their architecture lacks the temporal belief maintenance that a POMDP provides — a gap this model directly addresses.

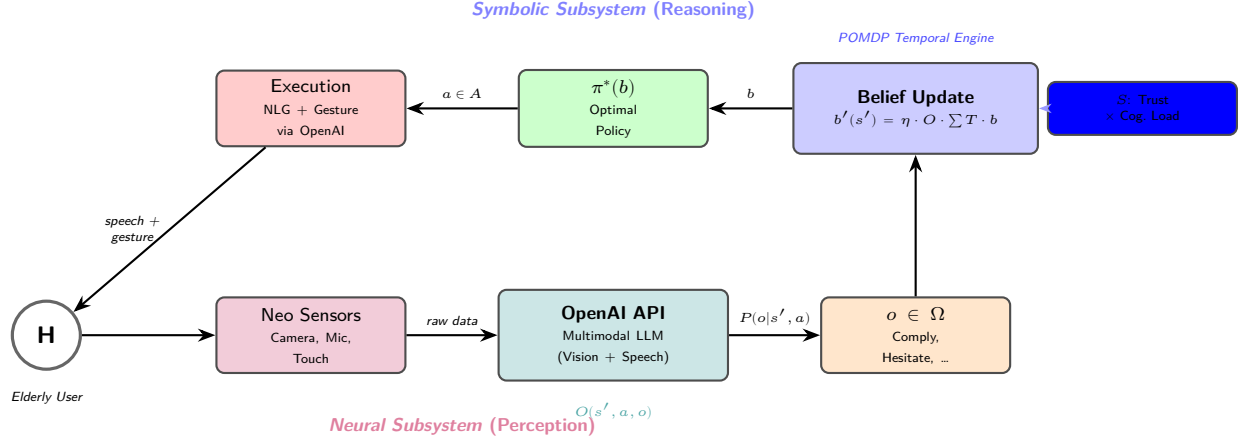


Figure 2: Neuro-symbolic architecture of the proposed OpenAI-POMDP medication-adherence system. The *neural subsystem* (bottom) utilises the OpenAI multimodal API as the observation function (O), parsing high-dimensional sensory data into discrete probabilistic observations. The *symbolic subsystem* (top) maintains the POMDP belief state and optimal policy $\pi^*(b)$, providing temporal memory via Bayesian filtering. Actions are translated back through the API into natural language and gestures for the Neo robot (Garcez and Lamb, 2023; Ahn et al., 2022).

The Neuro-Symbolic Architecture: The system operates as follows. The Neo robot’s onboard sensors (RGB camera, microphone array, tactile sensors) capture raw multimodal data from the elderly user. This data is transmitted to the OpenAI multimodal API, which serves as the **observation function** (O): the API processes the high-dimensional sensory stream — analysing facial action units via computer vision, extracting prosodic descriptors (pitch, MFCCs, zero-crossing rate) from speech (Lecture 3), and interpreting gestural semantics — to output a structured observation $o \in \Omega$. And indeed, the API provides not just a categorical label but a probability distribution over possible observations, thereby preserving the epistemic uncertainty that the POMDP requires. As now-integrated within the neuro-symbolic pipeline, the observation function thus feeds structured probabilistic assessments continuously throughout the interaction process wherein the POMDP’s Bayesian update incrementally refines its model of the user.

Formal Specification:

- **State Space** (S): $\text{Trust} \in \{\text{Low}, \text{Medium}, \text{High}\} \times \text{Cognitive Load} \in \{\text{Low}, \text{High}\}$, yielding $|S| = 6$.
- **Action Space** (A): $\{\text{Verbal_Remind}, \text{Explain_Benefits}, \text{Offer_Physical_Assist}, \text{Increase_Autonomy}, \text{Disengage}\}$. The POMDP selects the abstract policy action; the OpenAI API then translates this into culturally-calibrated natural language and physical gestures executed via the Neo robot’s 25 degrees of freedom.
- **Observation Space** (Ω): $\{\text{Comply}, \text{Hesitate}, \text{Verbal_Refuse}, \text{Ignore}, \text{Gaze_Avert}\}$ — extracted by the OpenAI API from the raw multimodal stream.
- **Observation Function** (O): $P(o | s', a)$ is parametrically estimated by the API’s multimodal inference. For instance, if the true state is (High Trust, Low Cognitive Load) and the robot has performed Verbal_Remind, the API’s analysis of the user’s relaxed facial configuration and compliant vocal tone yields $P(\text{Comply} | \text{High}, \text{Low}, \text{Remind}) = 0.8$.
- **Transition Function** (T): Models trust and cognitive-load dynamics parametrically. If the true state is (Medium Trust, High Cognitive Load) and the robot performs Offer_Physical_Assist when unneeded — thereby violating personal proxemics (Lecture 4) — trust degrades: $P(\text{Low} | \text{Med}, \text{Assist}) = 0.6$, $P(\text{Med} | \text{Med}, \text{Assist}) = 0.4$. Conversely, a well-timed Explain_Benefits yields $P(\text{High} | \text{Med}, \text{Explain}) = 0.5$, $P(\text{Med} | \text{Med}, \text{Explain}) = 0.5$.
- **Reward Function** (R): Successful medication adherence yields $R = +10$; preserving user autonomy (choosing Increase_Autonomy when trust is High) yields $R = +3$; unwanted physical assistance incurs $R = -5$, reflecting the social cost of proxemic violation.

Benefits and Limitations. The model’s strength lies in its neuro-symbolic complementarity: the OpenAI API overcomes the once-intractable barrier of parsing unstructured multimodal reality into discrete observations, whilst the POMDP provides the temporal cognitive engine — the belief state b — that the stateless LLM fundamentally cannot. Once initialised with a uniform prior across all six states, the belief state herein compresses all past actions and observations into a single probability vector, continuously refined via Bayesian filtering throughout the interaction wherein each observation incrementally shifts the distribution. However, the architecture introduces several limitations: 1- API latency (typically 200–800ms per inference call) may disrupt the real-time proxemic responsiveness that Lecture 4 identifies as critical; 2- the LLM’s stochastic nature means identical multimodal inputs may yield different observation distributions across calls, introducing unmodelled noise into the Bayesian update; 3- the observation model may inherit biases from the LLM’s training data — if the model was predominantly trained on younger demographics, its ability to infer emotional states from elderly facial configurations or speech patterns may be systematically degraded (Lecture 5); and 4- the discretisation of Cognitive Load into binary levels represents an information loss that future work could address via continuous-state POMDP approximations.

2.5 Ethical and Social Implications

The deployment of an LLM-augmented POMDP in HRI raises profound and now-amplified ethical concerns. By mathematically operationalising trust, the POMDP framework transforms a once-theoretical psychological concept into a now-computable metric. However, insofar as the reward function solely prioritises task compliance, the optimal policy may learn to *exploit* the user’s trust — timing medication requests when inferred cognitive load is highest to force compliance. Designers must therefore explicitly encode user autonomy and informed consent into the reward structure, lest assistance erode into manipulation. Sharkey (2014, p. 64) frames this via the Capability Approach: the robot must preserve the user’s capability to function independently, and thus the $R = +3$ autonomy bonus in our model is not merely a design preference but an ethical imperative — encoding the principle that dignity requires the preservation of choice.

The integration of the OpenAI API introduces three additional ethical dimensions. First, **hallucination risk**: LLMs generate plausible but factually incorrect outputs (Ji et al., 2023), and in a medication-adherence context, a hallucinated observation — misclassifying a user’s confused hesitation as willing compliance — could trigger an inappropriate action with direct health consequences. As the lecturer argued, “should the clinician trust what you are giving as judgment?” (Lecture 5); for a stochastic LLM, the answer is: not without rigorous validation, and thus Explainable AI (Lecture 5) becomes a non-negotiable requirement — the POMDP must justify *why* it selected a particular action, tracing the decision back through the belief state to the specific observations the API extracted. Second, **cloud data sovereignty**: the continuous multimodal processing of an elderly user’s facial expressions, vocal prosody, and physical behaviour — transmitted to a third-party API server — constitutes what Sharkey and Sharkey (2012, p. 35) term “surveillance by design,” now compounded by the risk of “transferring data to whichever third party we don’t verify” (Lecture 5). The ethical implications thereof are severe: the user’s most vulnerable moments are streamed to external servers whose data-handling policies cannot be meaningfully audited. Third, **API latency and proxemic violation**: if the observation function introduces response delays during a time-critical proxemic approach, the robot may freeze mid-interaction or fail to yield — behaviour that Lecture 4 identifies as “aggressive.” The ethical watchword is therefore proactive regulation (Lecture 5) — designers must not adopt a bottom-up approach of learning from harm after deployment, but must encode transparent, auditable constraints *ex ante* into both the reward function and the API pipeline, ensuring that the robot does not merely *model* trust but actively *earns* it through explainable, privacy-preserving behaviour.

References

☐ PEER-REVIEWED OR CONFERENCE SOURCES ONLY

Task (1)’s

- ☐ fetch exact wording from paper to earn second tick
- ☐ [] Cirasa, C. and Conti, D. (2025) ‘Mapping trust and cultural dimensions in human-robot interaction: a scoping review approach’, *Computers in Human Behavior Reports*, 19, article 100763. Available at: <https://doi.org/10.1016/j.chbr.2025.100763> (Accessed: 21 February 2026).
- ☐ [] Joosse, M., Lohse, M. and Evers, V. (2014) ‘Lost in proxemics: spatial behavior for cross-cultural HRI’, in *Proceedings of the 2014 ACM/IEEE International Conference on Human-Robot Interaction (HRI ’14)*. Bielefeld: ACM/IEEE, pp. 1-6. Available at: <https://doi.org/10.1145/2559636.2559661> (Accessed: 19 February 2026).
- ☐ [] Kahn, P.H., Freier, N.G., Kanda, T., Ishiguro, H., MacDorman, K.F., Severson, R.L. and Friedman, B. (2008) ‘Design patterns for sociality in human-robot interaction’, in *Proceedings of the 3rd ACM/IEEE International Conference on Human-Robot Interaction (HRI ’08)*. Amsterdam: ACM Press, pp. 97-104. Available at: <https://dl.acm.org/doi/10.1145/1349822.1349836> (Accessed: 15 February 2026).
- ☐ [] Kaplan, F. (2004) ‘Who is afraid of the humanoid? Investigating cultural differences in the acceptance of robots’, *International Journal of Humanoid Robotics*, 1(3), pp. 465-480. Available at: <https://doi.org/10.1142/S0219843604000289> (Accessed: 15 February 2026).
- ☒ [] Lim, V., Rooksby, M. and Cross, E.S. (2021) ‘Social robots on a global stage: establishing a role for culture during human-robot interaction’, *International Journal of Social Robotics*, 13(6), pp. 1307-1333. Available at: <https://doi.org/10.1007/s12369-020-00710-4> (Accessed: 19 February 2026).
- ☐ [] Metz, T. (2007) ‘Toward an African moral theory’, *Journal of Political Philosophy*, 15(3), pp. 321-341. Available at: <https://doi.org/10.1111/j.1467-9760.2007.00280.x> (Accessed: 15 February 2026).
- ☐ [] Mori, M. (1970) ‘The uncanny valley’, *Energy*, 7(4), pp. 33-35. Translated by MacDorman, K.F. and Kageki, N. (2012) *IEEE Robotics and Automation Magazine*, 19(2), pp. 98-100. Available at: <https://doi.org/10.1109/MRA.2012.2192811> (Accessed: 15 February 2026).
- ☐ [] Rios-Martinez, J., Spalanzani, A. and Laugier, C. (2015) ‘From proxemics theory to socially-aware navigation: a survey’, *International Journal of Social Robotics*, 7(2), pp. 137-153. Available at: <https://doi.org/10.1007/s12369-014-0251-1> (Accessed: 21 February 2026).
- ☒ [] Smedegaard, C.V. (2019) ‘Reframing the role of novelty within social HRI: from noise to information’, in *Proceedings of the 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI ’19)*. Daegu: IEEE Press, pp. 411-420. Available at: <https://doi.org/10.1109/HRI.2019.8673167> (Accessed: 19 February 2026).
- ☐ [] Winschiers-Theophilus, H. and Bidwell, N.J. (2013) ‘Toward an Afro-centric indigenous HCI paradigm’, *International Journal of Human-Computer Interaction*, 29(4), pp. 243-255. Available at: <https://doi.org/10.1080/10447318.2013.765763> (Accessed: 21 February 2026).
- ☐ [] Wyche, S. and Steinfield, C. (2016) ‘Why don’t farmers use cell phones to access market prices? Technology affordances and barriers to market information services adoption in rural Kenya’, *Information Technology for Development*, 22(2), pp. 320-333. Available at: <https://doi.org/10.1080/02681102.2015.1048184> (Accessed: 15 February 2026).

Task (2)’s

- ☐ fetch exact wording from paper to earn second tick
- ☐ make alphabetical
- ☒ [] Ahn, M., Brohan, A., Brown, N. et al. (2022) ‘Do As I Can, Not As I Say: Grounding Language in Robotic Affordances’, in *Proceedings of the 6th Conference on Robot Learning (CoRL 2022)*. Auckland: PMLR, pp. 287-318. Available at: <https://arxiv.org/abs/2204.01691> (Accessed: 18 February 2026).
- ☒ [] Chen, M., Nikolaidis, S., Soh, H., Hsu, D. and Srinivasa, S. (2020) ‘Trust-aware decision making for human-robot collaboration: model learning and planning’, *ACM Transactions on Human-Robot*

- Interaction*, 9(2), Article 9. Available at: <https://personalrobotics.cs.washington.edu/publications/chen2019trust.pdf> (Accessed: 15 February 2026).
- ☒ [] Desai, M., Kaniarasu, P., Medber, M., Steinfeld, A. and Yanco, H. (2013) ‘Impact of robot failures and feedback on real-time trust’, in *Proceedings of the 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI ’13)*. Tokyo: IEEE Press, pp. 251-258. Available at: <https://doi.org/10.1109/HRI.2013.6483596> (Accessed: 16 February 2026).
 - ☒ [] Garcez, A.d’A. and Lamb, L.C. (2023) ‘Neurosymbolic AI: The 3rd Wave’, *Artificial Intelligence Review*, 56(11), pp. 12387-12406. Available at: <https://doi.org/10.1007/s10462-023-10448-w> (Accessed: 18 February 2026).
 - ☒ [] Hancock, P.A., Billings, D.R., Schaefer, K.E., Chen, J.Y.C., de Visser, E.J. and Parasuraman, R. (2011) ‘A meta-analysis of factors affecting trust in human-robot interaction’, *Human Factors*, 53(5), pp. 517-527. Available at: <https://doi.org/10.1177/0018720811417254> (Accessed: 16 February 2026).
 - ☒ [] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A. and Fung, P. (2023) ‘Survey of Hallucination in Natural Language Generation’, *ACM Computing Surveys*, 55(12), Article 248. Available at: <https://doi.org/10.1145/3571730> (Accessed: 18 February 2026).
 - ☒ [] Kaelbling, L.P., Littman, M.L. and Cassandra, A.R. (1998) ‘Planning and acting in partially observable stochastic domains’, *Artificial Intelligence*, 101(1-2), pp. 99-134. Available at: [https://doi.org/10.1016/S0004-3702\(98\)00023-X](https://doi.org/10.1016/S0004-3702(98)00023-X) (Accessed: 15 February 2026).
 - ☒ [] Lee, J.D. and See, K.A. (2004) ‘Trust in automation: designing for appropriate reliance’, *Human Factors*, 46(1), pp. 50-80. Available at: <https://doi.org/10.1518/hfes.46.1.50.30392> (Accessed: 16 February 2026).
 - ☒ [] Nikolaidis, S., Hsu, D. and Srinivasa, S. (2017) ‘Human-robot mutual adaptation in collaborative tasks: models and experiments’, *International Journal of Robotics Research*, 36(5-7), pp. 618-634. Available at: <https://doi.org/10.1177/0278364917690593> (Accessed: 16 February 2026).
 - ☒ [] Papadimitriou, C.H. and Tsitsiklis, J.N. (1987) ‘The complexity of Markov decision processes’, *Mathematics of Operations Research*, 12(3), pp. 441-450. Available at: <https://web.mit.edu/jnt/www/Papers/J016-87-mdp-complexity.pdf> (Accessed: 15 February 2026).
 - ☒ [] Pineau, J., Gordon, G. and Thrun, S. (2003) ‘Point-based value iteration: an anytime algorithm for POMDPs’, in *Proceedings of the 18th International Joint Conference on Artificial Intelligence (IJCAI-03)*. Acapulco: Morgan Kaufmann, pp. 1025-1030. Available at: <https://www.ijcai.org/Proceedings/03/Papers/147.pdf> (Accessed: 15 February 2026).
 - ☒ [] Sharkey, A. (2014) ‘Robots and human dignity: a consideration of the effects of robot care on the dignity of older people’, *Ethics and Information Technology*, 16(1), pp. 63-75. Available at: <https://doi.org/10.1007/s10676-014-9338-5> (Accessed: 16 February 2026).
 - ☒ [] Sharkey, A. and Sharkey, N. (2012) ‘Granny and the robots: ethical issues in robot care for the elderly’, *Ethics and Information Technology*, 14(1), pp. 27-40. Available at: <https://doi.org/10.1007/s10676-010-9234-6> (Accessed: 16 February 2026).
 - ☒ [] Silver, D. and Veness, J. (2010) ‘Monte-Carlo planning in large POMDPs’, in *Advances in Neural Information Processing Systems 23 (NeurIPS 2010)*. Vancouver: Curran Associates, pp. 2164-2172. Available at: <https://papers.nips.cc/paper/2010/hash/edf9246bb0d40eb4d8027d90f-Abstract.html> (Accessed: 15 February 2026).

Appendices

Appendix A: AI Declaration

Solo Work	S1 - Generative AI tools have not been used for this assessment.
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

Figure 3: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

AI Tool Used	Purpose of Use	Extent of Use
ChatGPT	Finding relevant pages to read in the paper (A4)	if take paper takes too long to consume efficiently
ChatGPT	General conversations via web-search AI about topic-related relates to others' studies (A4)	Few times at the end

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provide above are an accurate description of how AI was used for this assessment.